# SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

## Office Visitor and Task Management System (OVTMS)

**Version:** 1.0
**Document Type:** Official System Documentation
**Prepared For:** Academic

# Document Control

| Item | Description |
|---|---|
| Document Title | Software Requirements Specification |
| System Name | Office Visitor and Task Management System (OVTMS) |
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# 1. INTRODUCTION

## 1.1 System Overview

The **Office Visitor and Task Management System (OVTMS)** is a web-based information system developed to automate office visitor registration, internal task coordination, follow-up management, staff monitoring, and reporting.

The system integrates:

- Visitor Management
- Task Lifecycle Management
- Follow-up Processing
- Notification Automation
- Dashboard Monitoring
- Report Generation

The system supports role-based operations for:

- Administrators
- Supervisors
- Staff
- Receptionists

## 1.2 Problem Statement

Traditional office visitor handling and task tracking often rely on manual records and fragmented communication which leads to:

- Lost records
- Delayed task processing
- Poor follow-up tracking
- Limited accountability
- Weak reporting mechanisms

OVTMS addresses these challenges through automation and centralized management.

# 2. PURPOSE OF THE SYSTEM

The system shall:

- Register and manage office visitors.
- Assign and monitor tasks.
- Track follow-up activities.
- Support supervision and approvals.
- Generate operational reports.
- Provide dashboards and alerts.
- Improve accountability and workflow efficiency.

# 3. SCOPE

## 3.1 In Scope

The system includes:

- User authentication
- Role validation
- Visitor check-in/check-out
- Task creation and assignment
- Task progress tracking
- Staff performance monitoring
- Follow-up management
- Reports generation
- Email notifications
- Administrative controls

## 3.2 Out of Scope

The following are excluded:

- Payroll processing
- Accounting systems
- Biometric attendance
- Public external portals

# 4. USERS AND ROLES

## 4.1 Administrator

Administrators have full control over the system.

## Functions

### 4.1.1 Manage Users

**Purpose:** Manage all user accounts.

#### Capabilities

- Create users
- Edit users details
- Assign roles
- Activate/deactivate accounts
- Reset passwords
- Manage permissions

#### Inputs

- Name
- Username
- Email
- Role
- Department
- Password

#### Outputs

- User records
- Role assignments

### 4.1.2 View Reports

Administrator can:

- View visitor reports
- View task reports
- View staff performance reports

- Export PDF/Excel reports
- Print reports

### 4.1.3 System Settings

Configure:

- Notification rules
- Deadlines
- Security policies
- Permissions
- Report formats

## 4.2 Supervisor

Supervisors monitor tasks and staff.

### Functions

### 4.2.1 Review Tasks

Actions:

- View assigned tasks
- Review progress
- Approve completion
- Flag delays

### 4.2.2 Monitor Staff Performance

Metrics:

- Productivity rates
- Pending tasks
- Completion rates

- Overdue assignments
- Response times

### 4.2.3 Approve Follow-Ups

Supervisor can:

- Review follow-up requests
- Approve/reject actions
- Escalate unresolved issues

## 4.3 Staff

Operational users who perform assigned tasks.

### Modules

### 4.3.1 View Assigned Tasks

Displays:

- Task ID
- Priority
- Deadline
- Status
- Related visitor references

### 4.3.2 Update Task Progress

Task statuses:

- Not Started
- In Progress
- Waiting Review
- Completed

### 4.3.3 Record Follow-Up Actions

Examples:

- Calls made
- Meetings scheduled
- Requests processed

## 4.4 Receptionist

Handles visitor registration and movement.

### Modules

### Register Visitor

Captures:

- Visitor Name
- Contact
- Purpose of Visit
- Department
- Staff to Meet
- Date/Time

### Check In Visitor

Process:

- Verify appointment
- Record arrival
- Notify host staff

### Check Out Visitor

Captures:

- Departure time
- Visit outcome
- Notes

# 5. FUNCTIONAL REQUIREMENTS

# 5.1 Login and Authentication Module

## Description

Controls user access.

## Inputs

- Username
- Password

## Functional Requirements

The system shall:

1. Authenticate users.

2.Validate:

- Username
- Password
- Account status

3 .Redirect users by role:

- Administrator dashboard
- Supervisor dashboard
- Staff dashboard
- Receptionist dashboard

4. On invalid login:

- Display error
- Deny access
- Log failed attempt

# 5.2 Task Management Module

## Create Task

Fields:

- Task title
- Description
- Priority
- Deadline
- Assigned staff
- Department

## Assign Task

System shall:

- Select staff member
- Assign task
- Send notification
- Add task to queue

## Task Progress States

- Assigned
- Started
- Processing
- Awaiting Review
- Completed

## Overdue Check Logic

If:
Current Date > Deadline

Then:

- Trigger overdue notification
- Alert staff
- Alert supervisor
- Alert administrator

Else:

- Mark task completed when approved

# 5.3 Email Notification Component

Automated notifications triggered by:

- New task assignment
- Overdue tasks
- Visitor arrival
- Follow-up requests
- Reports availability

Recipients:

- Staff
- Supervisors
- Administrators

Notification Channels:

- Email
- Dashboard alerts

# 5.4 Main Dashboard Component

Dashboard Widgets:

## Total Visitors Today

Displays daily visitor count.

## Overdue Tasks Alert

Displays delayed tasks.

## Pending Tasks Count

Shows incomplete tasks.

## Recent Activities Feed

Displays:

- Visitor activity
- Task updates
- Follow-up events

### Completed Tasks Count

Displays completed task metrics.

## 5.5 Reports Module

### Report Types

- Daily Visitor Report
- Task Status Report
- Staff Performance Report

### Filters

Reports can be filtered by:

- Daily
- Weekly
- Monthly
- Custom Date Range

### Outputs

- Screen Preview
- PDF
- Excel
- Print Format

## 5.6 Visitor Management Module

### Capture Visitor Details

Validation Rules:

- Required fields mandatory
- Contact validation
- Purpose cannot be blank

## Assign Person to See

System shall:

- Select host staff
- Link visitor to host
- Notify host

## Record Visit Outcome

Possible outcomes:

- Resolved
- Follow-up needed
- Action requested

If follow-up required:
Send to Follow-Up Module.

# 5.7 Follow-Up Processing Module

## Create Follow-Up Record

Contains:

- Reference ID
- Issue
- Assigned action

- Due date

## Assign Follow-Up

May assign to:

- Staff
- Supervisor
- Department

## Submit For Review

Reviewed by:

- Supervisor
- Administrator

# 5.8 Logout Component

The system shall:

- Destroy session
- Clear tokens
- Redirect to login page

# 6. DATABASE REQUIREMENTS

## Core Entities

## Users Table

| Field | Type | Key | Notes |
|---|---|---|---|
| id | BIGINT UNSIGNED | PK | Auto increment |
| name | VARCHAR(100) | | User full name |
| email | VARCHAR(150) | UNIQUE | Login email |
| role_id | BIGINT UNSIGNED | FK | References roles.id |
| password | VARCHAR(255) | | Hashed password |
| status | ENUM('active','inactive') | | Account state |
| created_at | TIMESTAMP | | Laravel default |
| updated_at | TIMESTAMP | | Laravel default |

# Roles Table

| Field | Type | Key |
|---|---|---|
| id | BIGINT UNSIGNED | PK |
| role_name | VARCHAR(50) | UNIQUE |

# Visitors Table

| Field | Type |
| --- | --- |
| id | BIGINT UNSIGNED |
| visitor_name | VARCHAR(100) |
| contact | VARCHAR(20) |
| purpose | TEXT |
| check_in | TIMESTAMP |
| check_out | TIMESTAMP NULL |
| host_staff_id | BIGINT UNSIGNED FK |
| visit_status | ENUM('checked_in','checked_out') |

# Tasks Table

| Field | Type |
| --- | --- |
| id | BIGINT UNSIGNED |
| title | VARCHAR(150) |
| description | TEXT |
| priority | ENUM('low','medium','high') |
| deadline | DATETIME |
| status | ENUM('pending','in_progress','completed') |
| assigned_to | BIGINT UNSIGNED FK |
| created_at | TIMESTAMP |
| updated_at | TIMESTAMP |

# FollowUps Table

| Field | Type |
| --- | --- |
| id | BIGINT UNSIGNED |
| task_id | BIGINT UNSIGNED FK |
| action | TEXT |
| review_status | ENUM('pending','approved','rejected') |
| created_at | TIMESTAMP |

# Notifications Table

| Field | Type |
| --- | --- |
| id | BIGINT UNSIGNED |
| recipient_id | BIGINT UNSIGNED FK |
| notification_type | VARCHAR(50) |
| message | TEXT |
| status | ENUM('read','unread') |
| created_at | TIMESTAMP |

# Reports Table

| Field | Type |
| --- | --- |
| id | BIGINT UNSIGNED |
| report_type | VARCHAR(50) |
| generated_by | BIGINT UNSIGNED FK |
| generated_date | TIMESTAMP |
| file_path | VARCHAR(255) NULL |

# 7. NON-FUNCTIONAL REQUIREMENTS

## Performance

The system shall:

- Respond in under 3 seconds
- Support concurrent users
- Process notifications in real time

## Reliability

- 99.9% uptime target
- Backup and recovery support
- Data consistency assurance

## Usability

- User-friendly dashboards
- Responsive interface
- Easy navigation

## Scalability

System shall support:

- Additional users
- Additional departments
- Growing visitor records

# 8. SECURITY REQUIREMENTS

## Authentication

The system shall support:

- Secure login
- Encrypted passwords
- Session timeout

## Authorization

Role-based access control shall ensure:

Receptionist cannot:

- Manage users
- Access admin reports

Staff cannot:

- Modify system settings
- Access administrator dashboard

Supervisors can:

- Review tasks
- Approve follow-ups

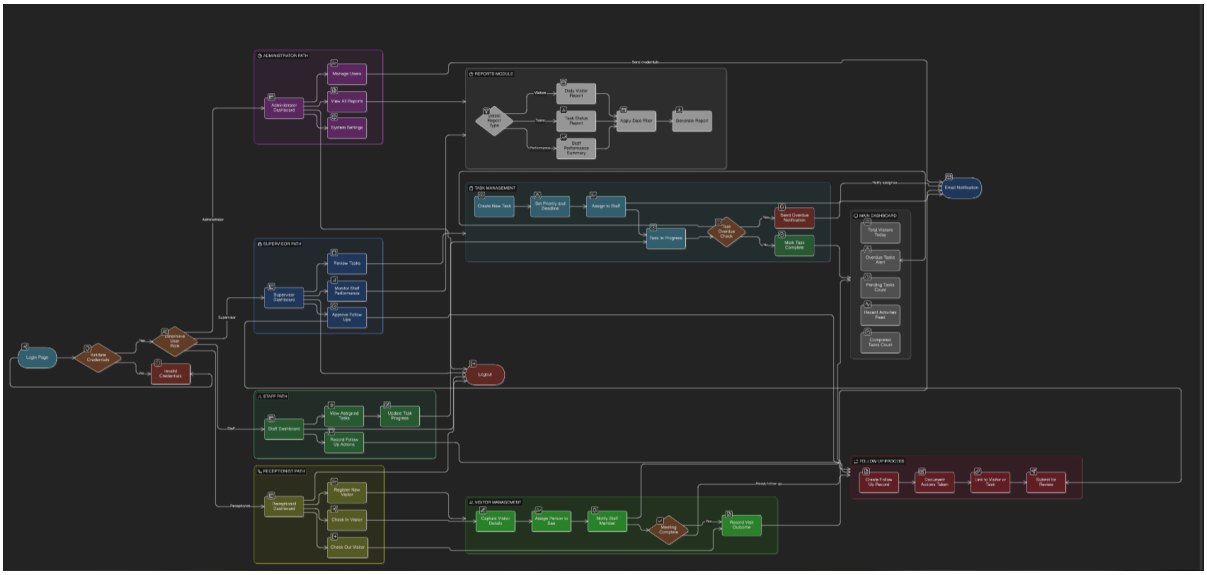
Administrators have:

- Full privileges

## Additional Security Controls

- Audit logs
- Activity tracking
- Failed login attempt logging
- CSRF protection
- Input validation
- SQL injection prevention
- Secure backups

# 9. SYSTEM ARCHITECTURE DIAGRAM




# 10. DEVELOPMENT RESOURCES AND TOOLS

## 10.1 Hardware Resources

- Personal Computer (PC)

- Stable Internet Connection

## 10.2 Software Tools

- Visual Studio Code
- Git
- GitHub (Version Control Repository)
- MySQL
- Draw.io (System Diagrams)
- Google Docs / Microsoft Word (Documentation)

## 10.3 Technologies Used

### Frontend

- HTML
- CSS
- Bootstrap
- JavaScript

### Backend

- Laravel (PHP Framework)

### Database

- MySQL

## 10.4 Development Skills Applied

- Web Development
- Database Design

- Version Control
- UI/UX Design
- System Analysis and Design

# 11. REFERENCES


1. IEEE 830 Software Requirements Specification Standard.
2. Laravel Documentation.
3. Bootstrap Documentation.
4. MySQL Documentation.
5. Git Documentation.
6. w3schools


# APPENDIX A: PROPOSED SYSTEM MODULES

Primary Modules:

1. Authentication Module
2. Visitor Management Module
3. Task Management Module
4. Follow-Up Module
5. Notifications Module
6. Reports Module
7. Administration Module

# APPENDIX B: FUTURE ENHANCEMENTS

Future versions may include:

- SMS notifications
- QR visitor passes
- Appointment booking
- Mobile support
- Analytics dashboard
- Biometric integration